

# IMO 2017 Solution Notes

Evan Chen 《陳誼廷》

18 October 2025

This is a compilation of solutions for the 2017 IMO. The ideas of the solution are a mix of my own work, the solutions provided by the competition organizers, and solutions found by the community. However, all the writing is maintained by me.

These notes will tend to be a bit more advanced and terse than the “official” solutions from the organizers. In particular, if a theorem or technique is not known to beginners but is still considered “standard”, then I often prefer to use this theory anyways, rather than try to work around or conceal it. For example, in geometry problems I typically use directed angles without further comment, rather than awkwardly work around configuration issues. Similarly, sentences like “let  $\mathbb{R}$  denote the set of real numbers” are typically omitted entirely.

Corrections and comments are welcome!

## Contents

|                                                     |          |
|-----------------------------------------------------|----------|
| <b>0 Problems</b>                                   | <b>2</b> |
| <b>1 Solutions to Day 1</b>                         | <b>3</b> |
| 1.1 IMO 2017/1, proposed by Stephan Wagner (SAF)    | 3        |
| 1.2 IMO 2017/2, proposed by Dorlir Ahmeti (ALB)     | 5        |
| 1.3 IMO 2017/3, proposed by Gerhard Woeginger (AUT) | 7        |
| <b>2 Solutions to Day 2</b>                         | <b>9</b> |
| 2.1 IMO 2017/4, proposed by Charles Leytem (LUX)    | 9        |
| 2.2 IMO 2017/5, proposed by Grigory Chelnokov (RUS) | 11       |
| 2.3 IMO 2017/6, proposed by John Berman (USA)       | 12       |

# IMO 2017解题笔记

陳誼廷 《Evan Chen》

2025年10月18日

这是2017年国际数学奥林匹克（IMO）的解题集。解题思路融合了我自己的工作、竞赛组织者提供的解法以及社区发现的解法。但所有写作均由我维护。

这些笔记在风格上可能比组织者提供的“官方”解法更进阶和简洁。特别是，如果一个定理或技巧对初学者不熟悉，但仍然被认为是“标准”的，我通常倾向于使用这个理论，而不是试图绕过或隐藏它。例如，在几何问题中，我通常会直接使用有向角，而不做进一步说明，而不是尴尬地绕过配置问题。类似地，“令  $\mathbb{R}$  表示实数集”这样的句子通常会被完全省略。

欢迎纠错和评论！

## 目录

|                                                |          |
|------------------------------------------------|----------|
| <b>0 问题</b>                                    | <b>2</b> |
| <b>1 Solutions to Day 1</b>                    | <b>3</b> |
| 1.1 IMO 2017/1, 由斯蒂芬·瓦格纳 (SAF) 提出.             | 3        |
| 1.2 IMO 2017/2, 由Dorlir Ahmeti (ALB) 提出.       | 5        |
| 1.3 IMO 2017/3, 由Gerhard Woeginger (AUT) 提出.   | 7        |
| <b>2 Solutions to Day 2</b>                    | <b>9</b> |
| 2.1 国际数学奥林匹克2017/4, 由查尔斯·莱泰姆（卢森堡）提出.           | 9        |
| 2.2 国际数学奥林匹克2017/5, 由Grigory Chelnokov（RUS）提出. | 11       |
| 2.3 国际数学奥林匹克2017/6, 由约翰·伯曼（美国）提出.              | 12       |

## §0 Problems

1. For each integer  $a_0 > 1$ , define the sequence  $a_0, a_1, a_2, \dots$ , by

$$a_{n+1} = \begin{cases} \sqrt{a_n} & \text{if } \sqrt{a_n} \text{ is an integer,} \\ a_n + 3 & \text{otherwise} \end{cases}$$

for each  $n \geq 0$ . Determine all values of  $a_0$  for which there is a number  $A$  such that  $a_n = A$  for infinitely many values of  $n$ .

2. Solve over  $\mathbb{R}$  the functional equation

$$f(f(x)f(y)) + f(x+y) = f(xy).$$

3. A hunter and an invisible rabbit play a game in the plane. The rabbit and hunter start at points  $A_0 = B_0$ . In the  $n$ th round of the game ( $n \geq 1$ ), three things occur in order:

- (i) The rabbit moves invisibly from  $A_{n-1}$  to a point  $A_n$  such that  $A_{n-1}A_n = 1$ .
- (ii) The hunter has a tracking device (e.g. dog) which reports an approximate location  $P_n$  of the rabbit, such that  $P_nA_n \leq 1$ .
- (iii) The hunter moves visibly from  $B_{n-1}$  to a point  $B_n$  such that  $B_{n-1}B_n = 1$ .

Let  $N = 10^9$ . Can the hunter guarantee that  $A_NB_N < 100$ ?

4. Let  $R$  and  $S$  be different points on a circle  $\Omega$  such that  $\overline{RS}$  is not a diameter. Let  $\ell$  be the tangent line to  $\Omega$  at  $R$ . Point  $T$  is such that  $S$  is the midpoint of  $\overline{RT}$ . Point  $J$  is chosen on minor arc  $RS$  of  $\Omega$  so that the circumcircle  $\Gamma$  of triangle  $JST$  intersects  $\ell$  at two distinct points. Let  $A$  be the common point of  $\Gamma$  and  $\ell$  closer to  $R$ . Line  $AJ$  meets  $\Omega$  again at  $K$ . Prove that line  $KT$  is tangent to  $\Gamma$ .

5. Fix  $N \geq 1$ . A collection of  $N(N+1)$  soccer players of distinct heights stand in a row. Sir Alex Song wishes to remove  $N(N-1)$  players from this row to obtain a new row of  $2N$  players in which the following  $N$  conditions hold: no one stands between the two tallest players, no one stands between the third and fourth tallest players,  $\dots$ , no one stands between the two shortest players. Prove that this is possible.

6. An *irreducible lattice point* is an ordered pair of integers  $(x, y)$  satisfying  $\gcd(x, y) = 1$ . Prove that if  $S$  is a finite set of irreducible lattice points then there exists a nonconstant *homogeneous* polynomial  $f(x, y)$  with integer coefficients such that  $f(x, y) = 1$  for each  $(x, y) \in S$ .

## §0 问题

1. 对于每个整数  $a_0 > 1$ , 定义序列  $a_0, a_1, a_2, \dots$ , 由

$$a_{n+1} = \begin{cases} \sqrt{a_n} & \text{if } \sqrt{a_n} \text{ is an integer,} \\ a_n + 3 & \text{otherwise} \end{cases}$$

对于每个  $n \geq 0$ . 确定所有  $a_0$  的值, 使得存在一个数  $A$ , 使得  $a_n = A$  对于无限多个  $n$  的值成立。

2. 在  $\mathbb{R}$  上解函数方程

$$f(f(x)f(y)) + f(x+y) = f(xy).$$

3. 一个猎人和一个隐形兔子在平面内玩一个游戏。兔子和猎人在点  $A_0 = B_0$  开始。在游戏的第  $n$  轮 ( $n \geq 1$ ), 按顺序发生三件事:

- (i) 兔子从  $A_{n-1}$  无形地移动到点  $A_n$ , 使得  $A_{n-1}A_n = 1$ 。
- (ii) 猎人有一个追踪设备 (例如狗), 该设备报告兔子的近似位置  $P_n$ , 使得  $P_nA_n \leq 1$ 。
- (iii) 猎人从  $B_{n-1}$  可见地移动到点  $B_n$ , 使得  $B_{n-1}B_n = 1$ 。令  $N = 10^9$ 。猎人能保证  $A_NB_N < 100$  吗?

4. 令  $R$  和  $S$  是圆  $\Omega$  上的不同点, 使得  $RS$  不是直径。令  $\ell$  是圆  $\Omega$  在  $R$  处的切线。点  $T$  是这样的, 使得  $S$  是  $RT$  的中点。点  $J$  被选择在圆  $\Omega$  的劣弧  $RS$  上, 使得三角形  $JST$  的外接圆  $\Gamma$  与  $\ell$  在两个不同的点上相交。令  $A$  是  $\Gamma$  和  $\ell$  中更靠近  $R$  的公共点。直线  $AJ$  再次与圆  $\Omega$  相交于  $K$ 。证明直线  $KT$  与  $\Gamma$  相切。

5. Fix  $N \geq 1$ . 一组具有不同身高的  $N(N+1)$  足球运动员站成一排。亚历克斯·宋爵士希望从这排人中移除  $N(N-1)$  人, 以获得一个由  $2N$  人组成的新队列, 其中满足以下  $N$  条件: 两名最高运动员之间无人站立, 第三和第四高运动员之间无人站立,  $\dots$ , 两名最矮运动员之间无人站立。证明这是可能的。

6. 一个不可约格点是一个有序的整数对  $(x, y)$  满足  $\gcd(x, y) = 1$ 。证明如果  $S$  是一个有限的不变格点集, 那么存在一个非恒等齐次多项式  $f(x, y)$  具有整数系数, 使得  $f(x, y) = 1$  对于每个  $(x, y) \in S$ 。

## §1 Solutions to Day 1

### §1.1 IMO 2017/1, proposed by Stephan Wagner (SAF)

Available online at <https://aops.com/community/p8633268>.

#### Problem statement

For each integer  $a_0 > 1$ , define the sequence  $a_0, a_1, a_2, \dots$ , by

$$a_{n+1} = \begin{cases} \sqrt{a_n} & \text{if } \sqrt{a_n} \text{ is an integer,} \\ a_n + 3 & \text{otherwise} \end{cases}$$

for each  $n \geq 0$ . Determine all values of  $a_0$  for which there is a number  $A$  such that  $a_n = A$  for infinitely many values of  $n$ .

The answer is  $a_0 \equiv 0 \pmod{3}$  only.

¶ **First solution.** We first compute the minimal term of any sequence, periodic or not.

#### Lemma

Let  $c$  be the smallest term in  $a_n$ . Then either  $c \equiv 2 \pmod{3}$  or  $c = 3$ .

*Proof.* Clearly  $c \neq 1, 4$ . Assume  $c \not\equiv 2 \pmod{3}$ . As  $c$  is not itself a square, the next perfect square after  $c$  in the sequence is one of  $(\lfloor \sqrt{c} \rfloor + 1)^2$ ,  $(\lfloor \sqrt{c} \rfloor + 2)^2$ , or  $(\lfloor \sqrt{c} \rfloor + 3)^2$ . So by minimality we require

$$c \leq \lfloor \sqrt{c} \rfloor + 3 \leq \sqrt{c} + 3$$

which requires  $c \leq 5$ . Since  $c \neq 1, 2, 4, 5$  we conclude  $c = 3$ .  $\square$

Now we split the problem into two cases:

- If  $a_0 \equiv 0 \pmod{3}$ , then all terms of the sequence are  $0 \pmod{3}$ . The smallest term of the sequence is thus 3 by the lemma and we have

$$3 \rightarrow 6 \rightarrow 9 \rightarrow 3$$

so  $A = 3$  works fine.

- If  $a_0 \not\equiv 0 \pmod{3}$ , then no term of the sequence is  $0 \pmod{3}$ , and so in particular 3 does not appear in the sequence. So the smallest term of the sequence is  $2 \pmod{3}$  by lemma. But since no squares are  $2 \pmod{3}$ , the sequence  $a_k$  grows without bound forever after, so no such  $A$  can exist.

Hence the answer is  $a_0 \equiv 0 \pmod{3}$  only.

## §1 Day 1的解法

### §1.1 IMO 2017/1, 由斯蒂芬·瓦格纳 (SAF) 提出

可在以下网址获取 <https://aops.com/community/p8633268>.

#### 问题描述

对于每个整数  $a_0 > 1$ , 定义序列  $a_0, a_1, a_2, \dots$ , 通过

$$a_{n+1} = \begin{cases} \sqrt{a_n} & \text{if } \sqrt{a_n} \text{ is an integer,} \\ a_n + 3 & \text{otherwise} \end{cases}$$

对于每个  $n \geq 0$ . 确定所有  $a_0$  的值, 使得存在一个数  $A$ , 使得  $a_n = A$  对于无穷多个  $n$  的值成立。

答案是  $a_0 \equiv 0 \pmod{3}$  仅。

¶ **首先解。** 我们首先计算任何序列的最小项, 无论是否周期性。

#### 引理

设  $c$  是  $a_n$  中的最小项。那么要么  $c \equiv 2 \pmod{3}$ , 要么  $c = 3$ 。

证明。显然  $c \neq 1, 4$ 。假设  $c \not\equiv 2 \pmod{3}$ 。由于  $c$  本身不是平方数, 序列中  $c$  之后下一个完全平方数是  $(\lfloor \sqrt{c} \rfloor + 1)^2$ ,  $(\lfloor \sqrt{c} \rfloor + 2)^2$ , 或  $(\lfloor \sqrt{c} \rfloor + 3)^2$  之一。所以根据最小性, 我们要求

$$c \leq \lfloor \sqrt{c} \rfloor + 3 \leq \sqrt{c} + 3$$

这需要  $c \leq 5$ 。由于  $c \neq 1, 2, 4, 5$  我们得出  $c = 3$ .  $\square$

现在我们将问题分为两种情况:

- 如果  $a_0 \equiv 0 \pmod{3}$ , 则序列的所有项都是  $0 \pmod{3}$ 。根据引理, 序列的最小项是 3, 因此我们有

$$3 \rightarrow 6 \rightarrow 9 \rightarrow 3$$

所以  $A = 3$  工作正常。

- 如果  $a_0 \not\equiv 0 \pmod{3}$ , 则序列中没有项是  $0 \pmod{3}$ , 因此特别是 3 不出现在序列中。根据引理, 序列的最小项是  $2 \pmod{3}$ 。但由于没有平方是  $2 \pmod{3}$ , 序列  $a_k$  在此之后无限增长, 因此没有这样的  $A$  可以存在。

因此答案仅为  $a_0 \equiv 0 \pmod{3}$ 。

¶ **Second solution.** We clean up the argument by proving the following lemma.

### Lemma

If  $a_n$  is constant modulo 3 and not 2 (mod 3), then  $a_n$  must eventually cycle in the form  $(m, m+3, m+6, \dots, m^2)$ , with no squares inside the cycle except  $m^2$ .

*Proof.* Observe that  $a_n$  must eventually hit a square, say  $a_k = c^2$ ; the next term is  $a_{k+1} = c$ . Then it is forever impossible to exceed  $c^2$  again, by what is essentially discrete intermediate value theorem. Indeed, suppose  $a_\ell > c^2$  and take  $\ell > k$  minimal (in particular  $a_\ell \neq \sqrt{a_{\ell-1}}$ ). Thus  $a_{\ell-1} \in \{c^2 - 2, c^2 - 1, c^2\}$  and thus for modulo 3 reasons we have  $a_{\ell-1} = c^2$ . But that should imply  $a_\ell = c < c^2$ , contradiction.

We therefore conclude  $\sup\{a_n, a_{n+1}, \dots\}$  is a decreasing integer sequence in  $n$ . It must eventually stabilize, say at  $m^2$ . Now we can't hit a square between  $m$  and  $m^2$ , and so we are done.  $\square$

Now, we contend that all  $a_0 \equiv 0 \pmod{3}$  work. Indeed, for such  $a_0$  we have  $a_n \equiv 0 \pmod{3}$  for all  $n$ , so the lemma implies that the problem statement is valid.

Next, we observe that if  $a_i \equiv 2 \pmod{3}$ , then the sequence grows without bound afterwards since no squares are 2 (mod 3). In particular, if  $a_0 \equiv 2 \pmod{3}$  the answer is no.

Finally, we claim that if  $a_0 \equiv 1 \pmod{3}$ , then eventually some term is 2 (mod 3). Assume for contradiction this is not so; then  $a_n \equiv 1 \pmod{3}$  must hold forever, and the lemma applies to give us a cycle of the form  $(m, m+3, \dots, m^2)$  where  $m \equiv 1 \pmod{3}$ . In particular  $m \geq 4$  and

$$m \leq (m-2)^2 < m^2$$

but  $(m-2)^2 \equiv 1 \pmod{3}$  which is a contradiction.

¶ **第二种解法。** 我们通过证明以下引理来清理论证。

### 引理

如果  $a_n$  模3是常数且不是2 (mod 3), 那么  $a_n$  最终必须在形式  $(m, m+3, m+6, \dots, m^2)$  的循环中, 循环内除  $m^2$  外没有其他平方数。

证明。观察到  $a_n$  最终会碰到一个平方数, 比如  $a_k = c^2$ ; 下一个数是  $a_{k+1} = c$ 。然后, 根据本质上的离散中值定理, 永远不可能再次超过  $c^2$ 。确实, 假设  $a_\ell > c^2$  并取  $\ell > k$  最小 (特别是  $a_\ell \neq \sqrt{a_{\ell-1}}$ )。因此  $a_{\ell-1} \in \{c^2 - 2, c^2 - 1, c^2\}$ , 并且因此对于模3的原因我们有  $a_{\ell-1} = c^2$ 。但这应该意味着  $a_\ell = c < c^2$ , 矛盾。

因此, 我们得出结论  $\sup\{a_n, a_{n+1}, \dots\}$  是一个在  $n$  中递减的整数序列。它最终必须稳定下来, 比如说在  $m^2$ 。现在我们不能在  $m$  和  $m^2$  之间得到一个平方数, 所以我们完成了证明。  $\square$

现在, 我们断言所有  $a_0 \equiv 0 \pmod{3}$  都有效。确实, 对于这样的  $a_0$ , 我们有  $a_n \equiv 0 \pmod{3}$  对于所有  $n$ , 因此引理意味着问题描述是有效的。

接下来, 我们观察到如果  $a_i \equiv 2 \pmod{3}$ , 那么序列之后会无界增长, 因为没有平方数是 2 (mod 3)。特别是, 如果  $a_0 \equiv 2 \pmod{3}$ , 答案就是否定的。

最后, 我们断言如果  $a_0 \equiv 1 \pmod{3}$ , 那么最终某个项是 2 (mod 3)。假设反证这个命题不成立; 那么  $a_n \equiv 1 \pmod{3}$  必须永远成立, 并且引理适用于给出一个形式为  $(m, m+3, \dots, m^2)$  的循环, 其中  $m \equiv 1 \pmod{3}$ 。特别是  $m \geq 4$  和

$$m \leq (m-2)^2 < m^2$$

但是  $(m-2)^2 \equiv 1 \pmod{3}$ , 这矛盾。

**§1.2 IMO 2017/2, proposed by Dorlir Ahmeti (ALB)**Available online at <https://aops.com/community/p8633190>.**Problem statement**Solve over  $\mathbb{R}$  the functional equation

$$f(f(x)f(y)) + f(x+y) = f(xy).$$

The only solutions are  $f(x) = 0$ ,  $f(x) = x - 1$  and  $f(x) = 1 - x$ , which clearly work.  
Note that

- If  $f$  is a solution, so is  $-f$ .
- Moreover, if  $f(0) = 0$  then setting  $y = 0$  gives  $f \equiv 0$ . So henceforth we assume  $f(0) > 0$ .

**Claim —** We have  $f(z) = 0 \iff z = 1$ . Also,  $f(0) = 1$  and  $f(1) = 0$ .

*Proof.* For the forwards direction, if  $f(z) = 0$  and  $z \neq 1$  one may put  $(x, y) = (z, z(z-1)^{-1})$  (so that  $x+y = xy$ ) we deduce  $f(0) = 0$  which is a contradiction.

For the reverse,  $f(f(0)^2) = 0$  by setting  $x = y = 0$ , and use the previous part. We also conclude  $f(1) = 0$ ,  $f(0) = 1$ .  $\square$

**Claim —** If  $f$  is injective, we are done.

*Proof.* Setting  $y = 0$  in the original equation gives  $f(f(x)) = 1 - f(x)$ . We apply this three times on the expression  $f^3(x)$ :

$$f(1 - f(x)) = f(f(f(x))) = 1 - f(f(x)) = f(x).$$

Hence  $1 - f(x) = x$  or  $f(x) = 1 - x$ .  $\square$

**Remark.** The result  $f(f(x)) + f(x) = 1$  also implies that surjectivity would solve the problem.

**Claim —**  $f$  is injective.

*Proof.* Setting  $y = 1$  in the original equation gives  $f(x+1) = f(x) - 1$ , and by induction

$$f(x+n) = f(x) - n. \quad (1)$$

Assume now  $f(a) = f(b)$ . By using (1) we may shift  $a$  and  $b$  to be large enough that we may find  $x$  and  $y$  obeying  $x+y = a+1$ ,  $xy = b$ . Setting these gives

$$\begin{aligned} f(f(x)f(y)) &= f(xy) - f(x+y) = f(b) - f(a+1) \\ &= f(b) + 1 - f(a) = 1 \end{aligned}$$

**§1.2 国际数学奥林匹克2017/2, 由Dorlir Ahmeti (ALB)提出**可在网上获取 <https://aops.com/community/p8633190>。**问题描述**求解函数方程 over  $\mathbb{R}$ 

$$f(f(x)f(y)) + f(x+y) = f(xy).$$

唯一解为  $f(x) = 0$ ,  $f(x) = x - 1$  和  $f(x) = 1 - x$ , 它们显然成立。注意

- 如果  $f$  是解, 那么  $-f$  也是解。
- 此外, 如果  $f(0) = 0$ , 那么设置  $y = 0$  就得到  $f \equiv 0$ 。因此, 我们假设  $f(0) > 0$ 。

**命题 —** 我们有  $f(z) = 0 \iff z = 1$ 。此外,  $f(0) = 1$  和  $f(1) = 0$ 。

证明。对于正向方向, 如果  $f(z) = 0$  且  $z \neq 1$ , 那么可以设  $(x, y) = (z, z(z-1)^{-1})$  (以便  $x+y = xy$ ) 我们推导出  $f(0) = 0$  这与假设矛盾。  
且  $z \neq 1$ , 那么可以设  $(x, y) = (z, z(z-1)^{-1})$  (以便  $x+y = xy$ ) 我们推导出  $f(0) = 0$  这与假设矛盾。

对于反向,  $f(f(0)^2) = 0$  通过设置  $x = y = 0$ , 并使用前面的部分。我们还得出  $f(1) = 0$ ,  $f(0) = 1$ 。  $\square$

**命题 —** 如果  $f$  是单射, 我们就完成了。

证明。在原始方程中设置  $y = 0$  得到  $f(f(x)) = 1 - f(x)$ 。我们在表达式  $f^3(x)$  上应用这个三次:

$$f(1 - f(x)) = f(f(f(x))) = 1 - f(f(x)) = f(x).$$

因此  $1 - f(x) = x$  或  $f(x) = 1 - x$ 。  $\square$

**备注。** 结果  $f(f(x)) + f(x) = 1$  也意味着满射将解决该问题。

**命题 —**  $f$  是单射。

证明。在原始方程中设置  $y = 1$  得到  $f(x+1) = f(x) - 1$ , 并通过归纳法

$$f(x+n) = f(x) - n. \quad (1)$$

假设现在  $f(a) = f(b)$ 。通过使用 (1) 我们可以将  $a$  和  $b$  调整得足够大, 以便我们可以找到  $x$  和  $y$  满足  $x+y = a+1$ ,  $xy = b$ 。设置这些得到

$$\begin{aligned} f(f(x)f(y)) &= f(xy) - f(x+y) = f(b) - f(a+1) \\ &= f(b) + 1 - f(a) = 1 \end{aligned}$$

from which we conclude

$$f(f(x)f(y) + 1) = 0.$$

Hence by the first claim we have  $f(x)f(y) + 1 = 1$ , so  $f(x)f(y) = 0$ . Applying the first claim again gives  $1 \in \{x, y\}$ . But that implies  $a = b$ .  $\square$

**Remark.** Jessica Wan points out that for any  $a \neq b$ , at least one of  $a^2 > 4(b-1)$  and  $b^2 > 4(a-1)$  is true. So shifting via (1) is actually unnecessary for this proof.

**Remark.** One can solve the problem over  $\mathbb{Q}$  using only (1) and the easy parts. Indeed, that already implies  $f(n) = 1 - n$  for all  $n$ . Now we induct to show  $f(p/q) = 1 - p/q$  for all  $0 < p < q$  (on  $q$ ). By choosing  $x = 1 + p/q$ ,  $y = 1 + q/p$ , we cause  $xy = x + y$ , and hence  $0 = f(f(1 + p/q)f(1 + q/p))$  or  $1 = f(1 + p/q)f(1 + q/p)$ .

By induction we compute  $f(1 + q/p)$  and this gives  $f(p/q + 1) = f(p/q) - 1$ .

从中我们得出

$$f(f(x)f(y) + 1) = 0.$$

因此根据第一主张我们有  $f(x)f(y) + 1 = 1$ , 所以  $f(x)f(y) = 0$ 。再次应用第一主张得到  $1 \in \{x, y\}$ 。但这意味着  $a = b$ 。  $\square$

**注意。** 杰西卡·万指出对于任何  $a \neq b$ , 至少有一个  $a^2 > 4(b-1)$  和  $b^2 > 4(a-1)$  为真。所以通过 (1) 的转换实际上对这个证明是不必要的。

备注。可以在实数集  $\mathbb{Q}$  上解决此问题, 仅使用 (1) 和简单部分。实际上, 这已经意味着  $f(n) = 1 - n$  对所有  $n$  成立。现在我们通过归纳法来证明  $f(p/q) = 1 - p/q$  对所有  $0 < p < q$  (在  $q$  上)。通过选择  $x = 1 + p/q$ ,  $y = 1 + q/p$ , 我们导致  $xy = x + y$ , 因此  $0 = f(f(1 + p/q)f(1 + q/p))$  或  $1 = f(1 + p/q)f(1 + q/p)$ 。通过归纳法我们计算  $f(1 + q/p)$ , 这给出了  $f(p/q + 1) = f(p/q) - 1$ 。

## §1.3 IMO 2017/3, proposed by Gerhard Woeginger (AUT)

Available online at <https://aops.com/community/p8633324>.

## Problem statement

A hunter and an invisible rabbit play a game in the plane. The rabbit and hunter start at points  $A_0 = B_0$ . In the  $n$ th round of the game ( $n \geq 1$ ), three things occur in order:

- (i) The rabbit moves invisibly from  $A_{n-1}$  to a point  $A_n$  such that  $A_{n-1}A_n = 1$ .
- (ii) The hunter has a tracking device (e.g. dog) which reports an approximate location  $P_n$  of the rabbit, such that  $P_nA_n \leq 1$ .
- (iii) The hunter moves visibly from  $B_{n-1}$  to a point  $B_n$  such that  $B_{n-1}B_n = 1$ .

Let  $N = 10^9$ . Can the hunter guarantee that  $A_NB_N < 100$ ?

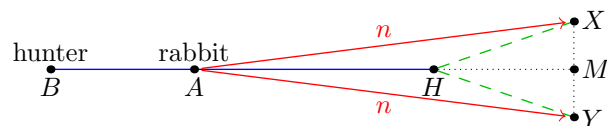
No, the hunter cannot. We will show how to increase the distance in the following way:

**Claim** — Suppose the rabbit is at a distance  $d \geq 1$  from the hunter at some point in time. Then it can increase its distance to at least  $\sqrt{d^2 + \frac{1}{2}}$  in  $4d$  steps regardless of what the hunter already knows about the rabbit.

*Proof.* Consider a positive integer  $n > d$ , to be chosen later. Let the hunter start at  $B$  and the rabbit at  $A$ , as shown. Let  $\ell$  denote line  $AB$ .

Now, we may assume the rabbit reveals its location  $A$ , so that all previous information becomes irrelevant.

The rabbit chooses two points  $X$  and  $Y$  symmetric about  $\ell$  such that  $XY = 2$  and  $AX = AY = n$ , as shown. The rabbit can then hop to either  $X$  or  $Y$ , ping the point  $P_n$  on the  $\ell$  each time. This takes  $n$  hops.



Now among all points  $H$  the hunter can go to,  $\min \max\{HX, HY\}$  is clearly minimized with  $H \in \ell$  by symmetry. So the hunter moves to a point  $H$  such that  $BH = n$  as well. In that case the new distance is  $HX = HY$ .

We now compute

$$\begin{aligned} HX^2 &= 1 + HM^2 = 1 + \left(\sqrt{AX^2 - 1} - AH\right)^2 \\ &= 1 + \left(\sqrt{n^2 - 1} - (n - d)\right)^2 \\ &\geq 1 + \left(\left(n - \frac{1}{n}\right) - (n - d)\right)^2 \\ &= 1 + (d - 1/n)^2 \end{aligned}$$

which exceeds  $d^2 + \frac{1}{2}$  whenever  $n \geq 4d$ .  $\square$

## §1.3 国际数学奥林匹克2017/3, 由 Gerhard Woeginger (AUT) 提出

可在以下网址获取 <https://aops.com/community/p8633324>.

## 问题描述

一名猎人和一只隐形兔子在平面内进行游戏。兔子和猎人都从点  $A_0 = B_0$  出发。在游戏的第  $n$  轮 ( $n \geq 1$ )，按顺序发生三件事：

- (i) 兔子从  $A_{n-1}$  无形地移动到点  $A_n$ ，使得  $A_{n-1}A_n = 1$ 。
- (ii) 猎人有一个追踪设备（例如狗），它会报告一个近似值位置  $P_n$  兔子，使得  $P_nA_n \leq 1$ 。
- (iii) 猎人可见地移动从  $B_{n-1}$  到一个点  $B_n$  使得  $B_{n-1}B_n = 1$ 。

令  $N = 10^9$ 。猎人能保证  $A_NB_N < 100$  吗？

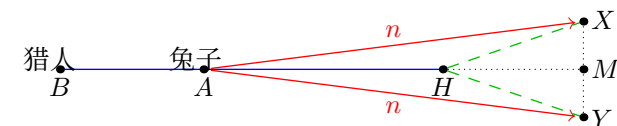
不能。我们将以下方式展示如何增加距离：

**命题** — 假设兔子在某个时间点与猎人的距离为  $d \geq 1$ 。那么它可以增加其距离至少  $\sqrt{d^2 + \frac{1}{2}}$  在  $4d$  步内，无论猎人已经知道关于兔子的什么信息。

*证明。* 考虑一个正整数  $n > d$ ，稍后选择。令猎人从  $B$  开始，兔子在  $A$ ，如图所示。令  $\ell$  表示线  $AB$ 。

现在，我们可以假设兔子透露其位置  $A$ ，以便所有先前信息 becomes 无关紧要。

兔子选择两个关于  $\ell$  对称的点  $X$  和  $Y$ ，如图所示。兔子可以跳到  $X$  或  $Y$ ，每次 ping  $\ell$  上的点  $P_n$ 。这需要  $n$  次跳跃。



现在，在猎人可以去的所有点  $H$  中， $\min \max\{HX, HY\}$  显然在  $H \in \ell$  处对称最小化。所以猎人移动到一个点  $H$ ，使得  $BH = n$  也成立。在这种情况下，新的距离是  $HX = HY$ 。

我们现在计算

$$\begin{aligned} HX^2 &= 1 + HM^2 = 1 + \left(\sqrt{AX^2 - 1} - AH\right)^2 \\ &= 1 + \left(\sqrt{n^2 - 1} - (n - d)\right)^2 \\ &\geq 1 + \left(\left(n - \frac{1}{n}\right) - (n - d)\right)^2 \\ &= 1 + (d - 1/n)^2 \end{aligned}$$

这超过了  $d^2 + \frac{1}{2}$ ，无论何时  $n \geq 4d$ 。  $\square$

In particular we can always take  $n = 400$  even very crudely; applying the lemma  $2 \cdot 100^2$  times, this gives a bound of  $400 \cdot 2 \cdot 100^2 < 10^9$ , as desired.

**Remark.** The step of revealing the location of the rabbit seems critical because as far as I am aware it is basically impossible to keep track of ping locations in the problem.

**Remark.** Reasons to believe the answer is “no”: the  $10^9$  constant, and also that “follow the last ping” is losing for the hunter.

**Remark.** I think there are roughly two ways you can approach the problem once you recognize the answer.

- (i) Try and control the location of the pings
- (ii) Abandon the notion of controlling possible locations, and try to increase the distance by a little bit, say from  $d$  to  $\sqrt{d^2 + \varepsilon}$ . This involves revealing the location of the rabbit before each iteration of several jumps.

I think it’s clear that the difficulty of my approach is realizing that (ii) is possible; once you do, the two-point approach is more or less the only one possible.  
My opinion is that (ii) is not that magical; as I said it was the first idea I had. But I am biased, because when I test-solved the problem at the IMO it was called “C5” and not “IMO3”; this effectively told me it was unlikely that the official solution was along the lines of (i), because otherwise it would have been placed much later in the shortlist.

特别是我们总是可以取  $n = 400$  , 即使非常粗糙; 应用引理 $2 \cdot 100^2$ 次, 这给出了一个界限  $400 \cdot 2 \cdot 100^2 < 10^9$ , 正如所希望的。

**注意。** 揭示兔子位置的一步似乎很关键, 因为我所知, 在问题中基本上不可能追踪ping位置。

**注意。** 相信答案为“否”的原因:  $10^9$  常数, 以及“跟随最后一个ping”对猎人是劣势的。

**注意。** 我认为一旦你认识到答案, 你可以大致有两种方法来处理这个问题。

- (i) 尝试控制ping的位置 (ii) 放弃控制可能位置的想法, 并尝试增加距离

稍微有点, 比如从  $d$  到  $\sqrt{d^2 + \varepsilon}$ 。这涉及到在每次跳跃的迭代之前揭示兔子的位置。

我认为很清楚, 我方法的困难在于认识到 (ii) 是可能的; 一旦你做到了, 两点法或多或少就是唯一可能的方法。  
我的观点是 (ii) 并不那么神奇; 正如我所说的, 这是我首先想到的想法。但我有偏见, 因为当我在国际数学奥林匹克上测试解决这个问题时, 它被称为“C5”, 而不是“IMO3”; 这实际上告诉我, 官方解法不太可能沿着 (i) 的方向发展, 否则它会被放在短名单的更后面。

## §2 Solutions to Day 2

### §2.1 IMO 2017/4, proposed by Charles Leytem (LUX)

Available online at <https://aops.com/community/p8639236>.

#### Problem statement

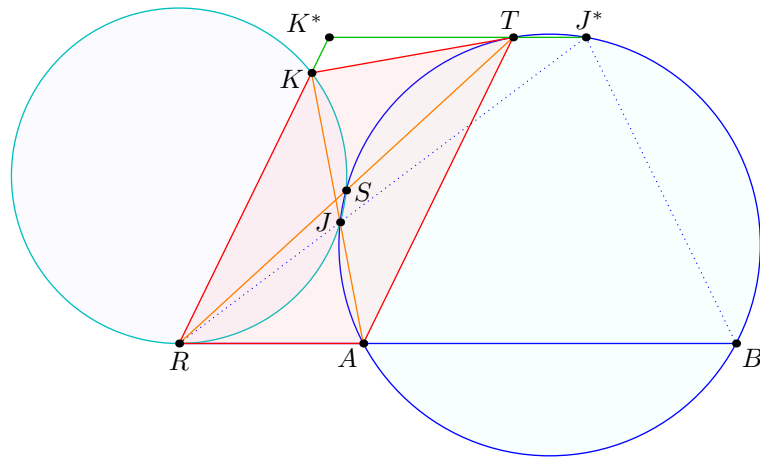
Let  $R$  and  $S$  be different points on a circle  $\Omega$  such that  $\overline{RS}$  is not a diameter. Let  $\ell$  be the tangent line to  $\Omega$  at  $R$ . Point  $T$  is such that  $S$  is the midpoint of  $\overline{RT}$ . Point  $J$  is chosen on minor arc  $RS$  of  $\Omega$  so that the circumcircle  $\Gamma$  of triangle  $JST$  intersects  $\ell$  at two distinct points. Let  $A$  be the common point of  $\Gamma$  and  $\ell$  closer to  $R$ . Line  $AJ$  meets  $\Omega$  again at  $K$ . Prove that line  $KT$  is tangent to  $\Gamma$ .

¶ **First solution (elementary).** First, note

$$\angle RKA = \angle RKJ = \angle RSJ = \angle TSJ = \angle TAJ = \angle TAK$$

so  $\overline{RK} \parallel \overline{AT}$ . Now,

- $\overline{RA}$  is tangent at  $R$  iff  $\triangle KRS \sim \triangle RTA$  (oppositely), because both equate to  $-\angle RKS = \angle SKR = \angle SRA = \angle TRA$ .
- Similarly,  $\overline{TK}$  is tangent at  $T$  iff  $\triangle ATS \sim \triangle TRK$ .
- The two similarities are equivalent because  $RS = ST$  the SAS gives  $KR \cdot TA = RS \cdot RT = TS \cdot TR$ .



**Remark.** The problem is actually symmetric with respect to two circles;  $\overline{RA}$  is tangent at  $R$  if and only if  $\overline{TK}$  at  $T$ .

¶ **Second solution (inversion).** Consider an inversion at  $R$  fixing the circumcircle  $\Gamma$  of  $TSJA$ . Then:

- $T$  and  $S$  swap,

## §2 Day 2的解法

### §2.1 IMO 2017/4, 由查尔斯·莱泰姆（卢森堡）提出

可在网上获取 <https://aops.com/community/p8639236>。

#### 问题描述

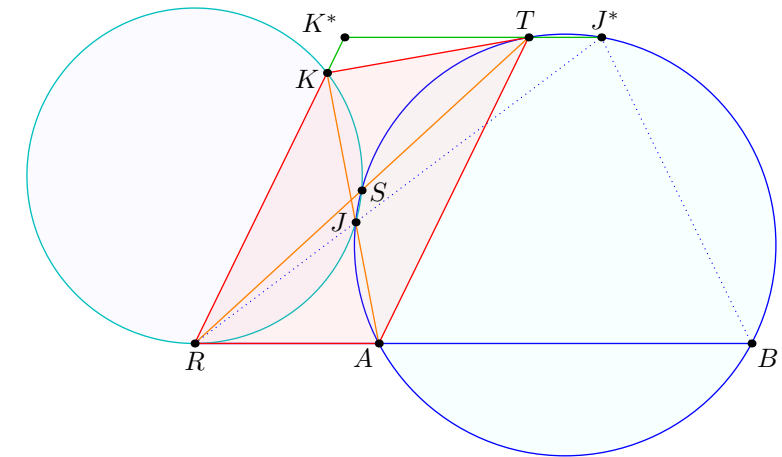
设  $R$  和  $S$  是圆  $\Omega$  上的两个不同点，且  $RS$  不是直径。设  $\ell$  是圆  $\Omega$  在  $R$  处的切线。点  $T$  满足  $S$  是  $RT$  的中点。点  $J$  在圆  $\Omega$  的劣弧  $RS$  上被选择，使得三角形  $JST$  的 circumcircle  $\Gamma$  与  $\ell$  相交于两个不同的点。设  $A$  是  $\Gamma$  和  $\ell$  中靠近  $R$  的共同点。直线  $AJ$  再次与  $\Omega$  相交于  $K$ 。证明直线  $KT$  与  $\Gamma$  相切。

¶ **第一个解（初等）。** 首先，注意

$$\angle RKA = \angle RKJ = \angle RSJ = \angle TSJ = \angle TAJ = \angle TAK$$

所以  $\overline{RK} \parallel \overline{AT}$ 。现在，

- $\overline{RA}$  在切线处， $R$  iff  $\triangle KRS \sim \triangle RTA$  (相反地)，因为它们都等于  $-\angle RKS = \angle SKR = \angle SRA = \angle TRA$ 。
- 类似地， $\overline{TK}$  在  $T$  处切线 iff  $\triangle ATS \sim \triangle TRK$ 。
- 这两个相似性是等价的，因为  $RS = ST$  SAS 给出  $KR \cdot TA = RS \cdot RT = TS \cdot TR$ 。



注意。这个问题实际上关于两个圆对称； $\overline{RA}$  相切  $R$  当且仅当  $\overline{TK}$  在  $T$ 。 at

¶ **第二个解法（反演）。** 考虑一个反演，固定外接圆  $\Omega$  的  $\Gamma$  于  $TSJA$ 。然后：

- $T$  和  $S$  交换，

- $A$  and  $B$  swap, where  $B$  is the second intersection of  $\ell$  with  $\Gamma$ .
- Circle  $\Omega$  inverts to the line through  $T$  parallel to  $\overline{RAB}$ , call it  $\ell$ .
- $J^*$  is the second intersection of  $\ell$  with  $\Gamma$ .
- $K^*$  is the intersection of  $\ell$  with the circumcircle of  $RBJ^*$ ; this implies  $RK^*J^*B$  is an isosceles trapezoid. In particular, one reads  $\overline{RK^*} \parallel \overline{AT}$  from this, hence  $RK^*TA$  is a parallelogram.

Thus we wish to show the circumcircle of  $RSK^*$  is tangent to  $\Gamma$ . But that follows from the final parallelogram observed:  $S$  is the center of the parallelogram since it is the midpoint of the diagonal.

**Remark.** This also implies  $RKTB$  is cyclic, from  $\overline{K^*SA}$  collinear. Moreover, quadrilateral  $KK^*TS$  is cyclic (by power of a point); this leads to the second official solution to the problem.

- $A$  and  $B$  交换, 其中  $B$  是第二个交点  $\ell$  with  $\Gamma$ .
- 圆  $\Omega$  映射到通过  $T$  平行于  $RAB$  的直线, 称为  $\ell$ 。
- $J^*$  是  $\ell$  与  $\Gamma$  的第二个交点。
- $K^*$  是  $\ell$  与  $RBJ^*$  的 circumcircle 的交点; 这意味着  $RK^*J^*B$  是一个等腰梯形。特别是, 由此可读出  $RK^* \parallel AT$ , 因此  $RK^*TA$  是一个平行四边形。

因此我们希望证明  $RSK^*$  的 circumcircle 与  $\Gamma$  相切。但这可以从观察到的最终平行四边形得出:  $S$  是平行四边形的中心, 因为它是对角线的中点。

**备注。** 这也意味着  $RKTB$  是循环的, 从  $K^*SA$  共线。此外, 四边形  $KK^*TS$  是循环的 (通过点的幂); 这导致了问题的第二个官方解。

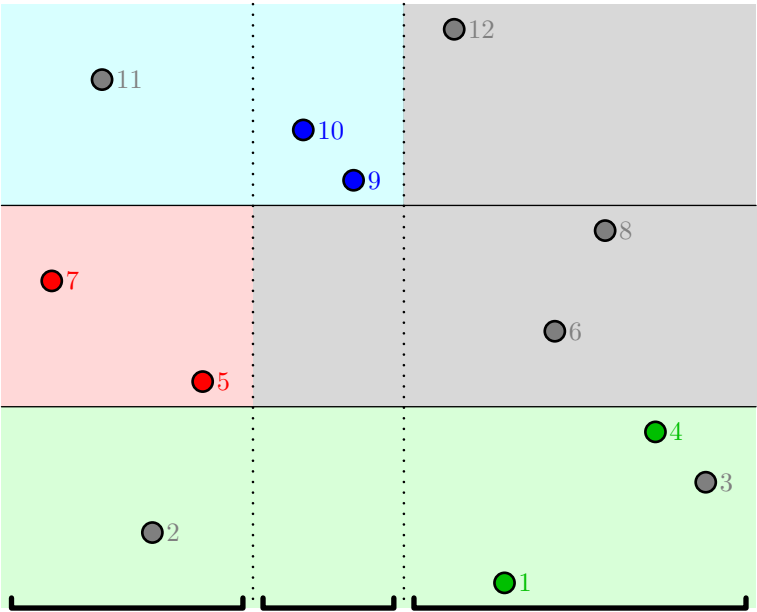
§2.2 IMO 2017/5, proposed by Grigory Chelnokov (RUS)

Available online at <https://aops.com/community/p8639240>.

Problem statement

Fix  $N \geq 1$ . A collection of  $N(N+1)$  soccer players of distinct heights stand in a row. Sir Alex Song wishes to remove  $N(N-1)$  players from this row to obtain a new row of  $2N$  players in which the following  $N$  conditions hold: no one stands between the two tallest players, no one stands between the third and fourth tallest players,  $\dots$ , no one stands between the two shortest players. Prove that this is possible.

Some opening remarks: **location and height are symmetric to each other**, if one thinks about this problem as permutation pattern avoidance. So while officially there are multiple solutions, they are basically isomorphic to one another, and I am not aware of any solution otherwise.



Take a partition of  $N$  groups in order by height:  $G_1 = \{1, \dots, N+1\}$ ,  $G_2 = \{N+2, \dots, 2N+2\}$ , and so on. We will pick two people from each group  $G_k$ . Scan from the left until we find two people in the same group  $G_k$ . Delete all people scanned and also everyone in  $G_k$ . All the groups still have at least  $N$  people left, so we can induct down with the non-deleted people; the chosen pair is to the far left anyways.

**Remark.** The important bit is to *scan by position* but *group by height*, and moreover not change the groups as we scan. Dually, one can have a solution which scans by height but groups by position.

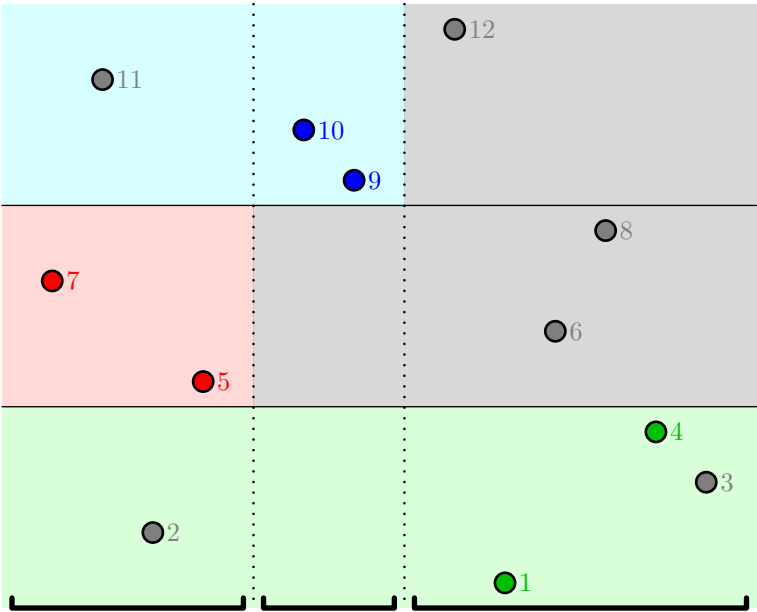
§ 2.2 IMO 2017/5, 由 Grigory Chelnokov (RUS) 提出

可在以下网址获取 <https://aops.com/community/p8639240>.

问题描述

Fix  $N \geq 1$ . A collection of  $N(N+1)$  足球运动员 of 不同身高 stand in a row. 亚历克斯·宋爵士 wishes to remove  $N(N-1)$  players from this row to obtain a new row of  $2N$  players in which the following  $N$  conditions hold: no one stands between the two 最高运动员, no one stands between the third and fourth tallest players,  $\dots$ , no one stands between the two 最矮运动员. Prove that this is possible.

Some opening remarks: **位置 and 身高是对称的**, if one thinks about this problem as 排列模式避免. So while officially there are 多种解法, they are basically 同构 to one another, and I am not aware of any solution otherwise.



按身高对  $N$  组进行划分:  $G_1 = \{1, \dots, N+1\}$ ,  $G_2 = \{N+2, \dots, 2N+2\}$ , 等等。我们将从每个组中挑选两个人。  
从左扫描, 直到找到同一组的两个人  $G_k$ 。删除所有扫描到的以及  $G_k$  中的人。所有组仍然至少有  $N$  人, 所以我们可以对未删除的人进行归纳下降; 选中的配对总是在最左边。

**备注。** 关键点是 按位置扫描 但 按身高分组, 而且扫描时不要改变分组。对偶地, 可以有 一种扫描按身高但分组按位置的解决方案。

## §2.3 IMO 2017/6, proposed by John Berman (USA)

Available online at <https://aops.com/community/p8639242>.

## Problem statement

An *irreducible lattice point* is an ordered pair of integers  $(x, y)$  satisfying  $\gcd(x, y) = 1$ . Prove that if  $S$  is a finite set of irreducible lattice points then there exists a nonconstant *homogeneous* polynomial  $f(x, y)$  with integer coefficients such that  $f(x, y) = 1$  for each  $(x, y) \in S$ .

We present two solutions.

¶ **First solution (Dan Carmon, Israel).** We prove the result by induction on  $|S|$ , with the base case being Bezout's Lemma ( $n = 1$ ). For the inductive step, suppose we want to add a given pair  $(a_{m+1}, b_{m+1})$  to  $\{(a_1, \dots, a_m), (b_1, \dots, b_m)\}$ .

**Claim (Standard)** — By a suitable linear transformation we may assume

$$(a_{m+1}, b_{m+1}) = (1, 0).$$

*Outline of proof.* It would be sufficient to show there exists a  $2 \times 2$  matrix  $T = \begin{bmatrix} u & v \\ s & t \end{bmatrix}$  with integer entries such that  $\det T = 1$  and  $T \cdot \begin{bmatrix} a_{m+1} \\ b_{m+1} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ . Then we could apply  $T$  to all the ordered pairs in  $S$  (viewed as column vectors); if  $f$  was a polynomial that works on the transformed ordered pairs, then  $f(ux + vy, sx + ty)$  works for the original ordered pairs. (Here, the condition that  $\det T = 1$  ensures that  $T^{-1}$  has integer entries, and hence that  $T$  maps irreducible lattice points to irreducible lattice points.)

To generate such a matrix  $T$ , choose  $T = \begin{bmatrix} u & v \\ -b_{m+1} & a_{m+1} \end{bmatrix}$  where  $u$  and  $v$  are chosen via Bézout lemma so that  $ua_{m+1} + vb_{m+1} = 1$ . This matrix  $T$  is rigged so that  $\det T = 1$  and the leftmost column of  $T^{-1}$  is  $\begin{bmatrix} a_{m+1} \\ b_{m+1} \end{bmatrix}$ .  $\square$

**Remark.** This transformation claim is not necessary to proceed; the solution below can be rewritten to avoid it with only cosmetic edits. However, it makes things a bit easier to read.

Let  $g(x, y)$  be a polynomial which works on the latter set. We claim we can choose the new polynomial  $f$  of the form

$$f(x, y) = g(x, y)^M - Cx^{\deg g \cdot M - m} \prod_{i=1}^m (b_i x - a_i y).$$

where  $C$  and  $M$  are integer parameters we may adjust.

Since  $f(a_i, b_i) = 1$  by construction we just need

$$1 = f(1, 0) = g(1, 0)^M - C \prod b_i.$$

If  $\prod b_i = 0$  we are done, since  $b_i = 0 \implies a_i = \pm 1$  in that case and so  $g(1, 0) = \pm 1$ , thus take  $M = 2$ . So it suffices to prove:

## §2.3 国际数学奥林匹克2017/6, proposed by 约翰·伯曼 (美国)

可在以下网址获取 <https://aops.com/community/p8639242>。

## 问题描述

一个不可约格点是满足 $\gcd(x, y) = 1$ 的整数有序对 $(x, y)$ 。证明如果  $S$  是一个有限个不可约格点的集合，那么存在一个非常数多项式  $f(x, y)$ ，其系数为整数，使得对于每个  $(x, y) \in S$ ，有  $f(x, y) = 1$ 。

我们给出两种解法。

¶ **第一个解法 (Dan Carmon, 以色列)。** 我们通过归纳法对  $|S|$  进行证明，基准情况是贝祖定理 ( $n = 1$ )。对于归纳步骤，假设我们要添加一个给定的对  $(a_{m+1}, b_{m+1})$  到  $\{(a_1, \dots, a_m), (b_1, \dots, b_m)\}$ 。

**命题 (标准)** — 通过适当的线性变换，我们可以假设

$$(a_{m+1}, b_{m+1}) = (1, 0).$$

**证明概要。** 如果能够证明存在一个  $2 \times 2$  矩阵  $T = \begin{bmatrix} u & v \\ s & t \end{bmatrix}$  具有整数元素，使得  $\det T = 1$  和  $T \cdot \begin{bmatrix} a_{m+1} \\ b_{m+1} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ 。那么我们可以将  $T$  应用于  $S$  中的所有有序对（将其视为列向量）；如果  $f$  是一个在变换后的有序对上工作的多项式，那么  $f(ux + vy, sx + ty)$  对原始有序对也适用。（在这里， $\det T = 1$  的条件确保  $T^{-1}$  具有整数元素，因此  $T$  将不可约格点映射到不可约格点。）

要生成这样的矩阵  $T$ ，选择  $T = \begin{bmatrix} u & v \\ -b_{m+1} & a_{m+1} \end{bmatrix}$ ，其中  $u$  和  $v$  是通过贝祖定理选择的，以便  $ua_{m+1} + vb_{m+1} = 1$ 。这个矩阵  $T$  被设置成  $\det T = 1$ ，并且  $T^{-1}$  的最左侧列是  $\begin{bmatrix} a_{m+1} \\ b_{m+1} \end{bmatrix}$ 。  $\square$

**注意。** 这个变换命题不是继续进行的必要条件；下面的解可以仅通过美观的修改来避免它。然而，它使阅读稍微容易一些。

设  $g(x, y)$  是作用于后一个集合的多项式。我们声称我们可以选择形式为  $f$  的新多项式。

$$f(x, y) = g(x, y)^M - Cx^{\deg g \cdot M - m} \prod_{i=1}^m (b_i x - a_i y).$$

其中  $C$  和  $M$  是我们调整的整数参数。

由于  $f(a_i, b_i) = 1$  是通过构造得到的，我们只需要

$$1 = f(1, 0) = g(1, 0)^M - C \prod b_i.$$

如果  $\prod b_i = 0$  我们完成了，因为  $b_i = 0 \implies a_i = \pm 1$  在这种情况下，所以  $g(1, 0) = \pm 1$ ，因此取  $M = 2$ 。所以足以证明：

**Claim** — We have  $\gcd(g(1, 0), b_i) = 1$  when  $b_i \neq 0$ .

*Proof.* Fix  $i$ . If  $b_i = 0$  then  $a_i = \pm 1$  and  $g(\pm 1, 0) = \pm 1$ . Otherwise know

$$1 = g(a_i, b_i) \equiv g(a_i, 0) \pmod{b_i}$$

and since the polynomial is homogeneous with  $\gcd(a_i, b_i) = 1$  it follows  $g(1, 0) \not\equiv 0 \pmod{b_i}$  as well.  $\square$

Then take  $M$  a large multiple of  $\varphi(\prod |b_i|)$  and we're done.

¶ **Second solution (Lagrange).** The main claim is that:

**Claim** — For every positive integer  $N$ , there is a homogeneous polynomial  $P(x, y)$  such that  $P(x, y) \equiv 1 \pmod{N}$  whenever  $\gcd(x, y) = 1$ .

(This claim is actually implied by the problem.)

*Proof.* For  $N = p^e$  a prime take  $(x^{p-1} + y^{p-1})^{\varphi(N)}$  when  $p$  is odd, and  $(x^2 + xy + y^2)^{\varphi(N)}$  for  $p = 2$ .

Now, if  $N$  is a product of primes, we can collate coefficient by coefficient using the Chinese remainder theorem.  $\square$

Let  $S = \{(a_i, b_i) \mid i = 1, \dots, m\}$ . We have the natural homogeneous “Lagrange polynomials”

$$L_k(x, y) := \prod_{i \neq k} (b_i x - a_i y)$$

Now let

$$N := \prod_k L_k(x_k, y_k)$$

and take  $P$  as in the claim. Then we can take a large power of  $P$ , and for each  $i$  subtract an appropriate multiple of  $L_i(x, y)$ ; that is, choose

$$f(x, y) = P(x, y)^C - \sum_i L_i(x, y) \cdot Q_i(x, y)$$

where  $C$  is large enough that  $C \deg P > \max_i \deg L_i$ , and  $Q_i(x, y)$  is any homogeneous polynomial of degree  $C \deg P - \deg L_i$  such that  $L_k(x_k, y_k) Q_k(x_k, y_k) = \frac{P(x_k, y_k)^{C-1}}{N}$ .  $L_k(x_k, y_k)$  (which is an integer).

**命题** — 我们有最大公约数  $(g(1, 0), b_i) = 1$  当  $b_i \neq 0$ 。

证明。固定  $i$ 。如果  $b_i = 0$  则  $a_i = \pm 1$  且  $g(\pm 1, 0) = \pm 1$ 。否则知道

$$1 = g(a_i, b_i) \equiv g(a_i, 0) \pmod{b_i}$$

并且由于多项式是齐次的,  $\gcd(a_i, b_i) = 1$  所以  $g(1, 0) \not\equiv 0 \pmod{b_i}$  同样成立。  $\square$

然后取  $M$  一个  $\varphi(\prod |b_i|)$  的大倍数, 就完成了。

¶ **第二种解法 (拉格朗日)**。主要命题是:

**命题** — 对于每一个正整数  $N$ , 存在一个齐次多项式  $P(x, y)$  使得  $P(x, y) \equiv 1 \pmod{N}$  当  $\gcd(x, y) = 1$  时。

(此命题实际上由问题隐含。)

证明。对于  $N = p^e$  一个素数取  $(x^{p-1} + y^{p-1})^{\varphi(N)}$  当  $p$  是奇数时, 和  $(x^2 + xy + y^2)^{\varphi(N)}$  当  $p = 2$ 。

现在, 如果  $N$  是素数的乘积, 我们可以使用中国剩余定理逐项合并系数。  $\square$

设  $S = \{(a_i, b_i) \mid i = 1, \dots, m\}$ 。我们有自然齐次的 “拉格朗日多项式”

$$L_k(x, y) := \prod_{i \neq k} (b_i x - a_i y)$$

现在让

$$N := \prod_k L_k(x_k, y_k)$$

并取  $P$  如命题中所述。然后我们可以取  $P$  的高次幂, 并对每个  $i$  减去适当的  $L_i$  的倍数  $(x, y)$ ; 即, 选择

$$f(x, y) = P(x, y)^C - \sum_i L_i(x, y) \cdot Q_i(x, y)$$

其中  $C$  足够大, 使得  $C \deg P > \max_i \deg L_i$ , 并且  $Q_i(x, y)$  是任何次数为

$C \deg P - \deg L_i$  的齐次多项式, 使得  $L_k(x_k, y_k) Q_k(x_k, y_k) = \frac{P(x_k, y_k)^{C-1}}{N}$ 。 (这是一个整数)。